Standard Operating Procedures for Final Year Project (BS CS/SE/AI/IT Program)

Department of Computer Science / Software Engineering

Faculty of Engineering & Computing

NATIONAL UNIVERSITY OF MODERN LANGUAGES
ISLAMABAD

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1. Introduction

The Final Year Project (FYP) is a 6-credit hour’s mandatory part of the BSSE/ BSCS/BSIT/BSAI program. The Final Year Project (FYP) aims to enable the students to practically implement all the theoretical knowledge. The FYP involves the proposal, design, and development of a real and substantial project related to software engineering, Computer Science, Information Technology, and Artificial Intelligence. It provides an opportunity for the students to crystallize their acquired professional competence in the form of a demonstrable software, simulation, or hardware product.

2. Roles and Responsibilities

The roles and responsibilities of departmental stakeholders are defined in the following subsections.

2.1. Role and Responsibilities of a Student

Every FYP student is responsible to:

1. Conduct regular meetings with the supervisor. Each student/ Group is required to conduct a biweekly meeting with the supervisor.
2. Follow the deadlines, i.e proposal submission, progress presentation, and final presentation
3. Maintain the meeting log (Appendix B) for every meeting. Provide the photocopy of the meeting log (after every meeting) to the supervisor and the project coordinator for the record.
4. Perform the assigned tasks regularly.
5. Seek guidance from the supervisor and explore online resources in case of any problem.
6. Present his/her work in front of the project evaluation committee according to the schedule.
7. Accommodate the changes recommended by the supervisor and/or project evaluation committee.
8. Report any grievance to HoD/departmental grievance committee.
9. The Proposal should map at least one SDG Goal.

2.2. Role and Responsibilities of a Faculty Member

Every faculty member is required to:

1. Submit two (or more) project ideas every semester. The project idea includes the project domain (e.g. web application), brief description, tools and technologies involved, team size required, and any additional information according to the template (Appendix A).
2. Guide and motivate FYP students whenever they seek guidance in general.

2.3. Role and Responsibilities of a Project Supervisor

A project supervisor is required to:

1. Conduct regular meetings with students under his/her supervision to discuss/evaluate their performance. At the end of each meeting, a meeting log (Appendix B) will be filled out by students and signed by the supervisor.
2. Motivate the students to participate in IGNITE funding. The supervisor conduct the mock presentations for IGNITE funding
3. Advise them for all activities related to FYP such as defining the project scope, selecting the appropriate tools and technology, and process model, etc.
4. Guide them in report writing according to the approved templates.
5. Review the writing drafts prepared from time to time for official submission.
6. Evaluate his/her students' performance at the end of the project (according to the defined rubric – Appendix F) or whenever required by the department.

2.4. Role and Responsibilities of Project Committee

The formation of the project committee is performed by HoD. The project committee is required to:

1. Evaluate project proposals, progress, final presentations, and reports according to the defined rubrics (Appendix C, D, and E) and schedule.
2. Provide critical feedback to students for the improvement of their projects.

2.5. Role and Responsibilities of Project Coordinator

One of the project committee members is assigned the role of Project Coordinator by HoD. The project coordinator is required to:

1. Manage all the activities related to the project proposal, progress, and final presentations.
2. Maintain all the relevant records of project approval, supervision, and evaluation.
3. Schedule the seminars and project evaluation activities according to the approved timeline.
4. Display the results after every evaluation activity.

3. Standard Operating Procedures

This section describes the standard operating procedures required to follow for the final year project.

3.1. Group Formation for FYP

1. Every project must be performed by a group of two to three students.
2. Students are free to form their groups according to their skill sets, interests, and the nature of the project.
3. If a single student wants to do FYP independently then he/she must need prior approval from HoD.

3.2. Project Supervision

1. Every faculty member may supervise a maximum of five projects at a time: including ongoing and new projects.
2. The project coordinator may ask students to change the supervisor (at the time of project proposal submission) to ensure the maximum number of projects allocated to any faculty member.

3.3. Project Timeline

1. The following timeline is defined for 6th, 7th, and 8th semester students to complete the project efficiently and smoothly.

Activities and timeline for 6^{th-} semester students		
Sr.	**Activity**	**Timeline**
1	Call for project ideas (for the faculty)	4^{th} week of the semester
2	Submission of project ideas (by the faculty)	6^{th} week of the semester
3	Display of project ideas	7^{th} week of the semester
4	Seminar on Project ideas	7^{th} week of the semester
5	Submission of project Ideas	10^{th} week of the semester
6	Presentation/evaluation of project Ideas	13^{th} week of the semester
7	Resubmission of project Ideas (in case of rejection)	14^{th} week of the semester
8	Presentation/evaluation of re-submitted project	15^{th} week of the semester
9	Display of accepted project Ideas	15^{th} week of the semester

Activities and timeline for 7^{th-} semester students		
Sr.	**Activity**	**Timeline**
1	Seminar on FYP Proposal Writing	2^{nd} week of the semester
2	Submission of the FYP Proposal document	4^{th} week of the semester
3	Evaluation of 40% Project Proposal Defense (Presentation)	4^{th} week of the semester
4	Seminar on FYP Report Writing (4 Chapters)	10^{th} week of the semester
5	Submission of the FYP Report (4 Chapters)	15^{th} week of the semester
6	Evaluation of 40% Project Progress Defense (Presentation)	15^{th} week of the semester

Activities and timeline for 8^{th} semester students		
Sr.	**Activity**	**Timeline**
1	Seminar on final report writing	4^{th} week of the semester
2	Submission of final report (hard and soft copy)	10^{th} week of the semester
3	Plagiarism check	11^{th} week of the semester
4	Evaluation of 100% project implementation	13^{th} week of the semester
5	Re-evaluation (if required)	14^{th} week of the semester
6	Resubmission of final report (2^{nd} submission for plagiarism check if required)	14^{th} week of the semester
7	Submission of hard binding report (final)	16^{th}week of the semester

3.4. Submission and Evaluation of Project Proposal

1. A student is eligible to take up a project if he/she has secured a minimum CGPA of 2.0 and his pending failed courses (Programming Subjects OOP,Programming fundemntals) in all semesters are not more than two.
2. Students may select the project from the list displayed by the department or propose their idea.
3. A formal project proposal prepared with the help of the supervisor and countersigned by him/her is submitted to the Project Coordinator according to the timeline defined by the department.
4. Students must have at least two meetings before the project proposal submission and the record must be maintained and presented at the time of presentation. Students are not allowed to present the proposal without the meeting log and 50% of the total marks allocated for this evaluation activity will be deducted. Students will be given another chance to present after providing the meeting log.
5. Students are required to present the project proposal in front of the Project Committee according to the schedule.
6. The project committee evaluates the project proposal according to the defined rubric (Appendix C).
7. The project proposal is either approved as it is, approved with modifications, or rejected.
8. In case of rejection, students have another chance to submit a new project proposal within the given timeline. 25% of the total allocated marks for this evaluation activity will be deducted. The second time rejection leads to failure and students need to submit the project proposal in the next semester
9. A project proposal approval report (result) is prepared and forwarded by the Project Coordinator to HoD within three days after presentations.
10. Results are displayed by the project coordinator after the formal approval.
11. Every approved project proposal will be submitted for the ignite funding (Ministry of information technology and telecom) whenever the funding call will be opened.

3.5. Submission and Evaluation of Project Progress

1. 40% of project work must be implemented for the presentation of project progress. In general, it is quantified based on the total number of modules / functional requirements to be implemented.

2. Students submit four chapters of the final project report countersigned by the supervisor according to the defined timeline.
3. Students need to submit the undertaking, countersigned by the supervisor, about the completion of 40% project work.
4. Students must show the meeting log at the time of presentation; otherwise, they will not be allowed to present the project progress. 50% of the total marks allocated for this evaluation activity will be deducted. Students will be given another chance to present after providing the meeting log.
5. The project committee evaluates and grades the project's progress according to the defined rubric (Appendix D).
6. Results are prepared and forwarded by the Project Coordinator to HoD within three days after presentations.
7. Results are displayed by the project coordinator after the formal approval.

3.6. Submission and Evaluation of the Final Project

1. At the time of the final presentation, a student can have only one pending/failed course from previous semesters. In that case, his/her result will not be declared until he clears failed course(s).
2. Students are required to submit the project completion certificate, countersigned by the supervisor. In case of less than 80% project completion or more than 18% similarity index of the final report, students are not allowed to present their project.
3. Students submit the final report countersigned by the supervisor according to the defined timeline. The same report will be forwarded to the project committee and checked for plagiarism.
4. The similarity index of the report must not be more than 18%. Students will be given two chances (maximum) to reduce the similarity index in case of exceeding the defined limit.
5. Students must show the meeting log at the time of presentation; otherwise, they will not be allowed to present the project progress. 50% of the total marks allocated for this evaluation activity will be deducted. Students will be given another chance to present after providing the meeting log.
6. The project committee evaluates and grades the final project according to the defined rubric (Appendix E).

7. In case of poor performance, students will be given a chance to re-demonstration (re-demo). The re-demo will be conducted after 15 days (maximum) from the date of the final presentation. If the project committee is not satisfied after the re-demo, students shall present their project in the next semester.
8. Results are prepared and forwarded by the Project Coordinator to HoD within three days after presentations.
9. Results are notified by the examination department after the formal approval.
10. A student will be responsible for fees or any other dues (for extra semesters) in case of failure/delay at any stage of the project.

3.7. Submission of Final Report

1. After the formal approval of the report from the department, students are required to submit three hard copies of the report to the project coordinator. These copies will be distributed to the department, library, and supervisor after being signed by the HoD and Dean.
2. The report must be hard bound in green for (SE)/black for (CS) color and the text must be embossed in silver.
3. The degree title along with the batch number, project title, and year of completion must be written on the spine of the hard binding.
4. A CD must be attached at the end of hard binding containing a certificate with original signatures.
5. The CD should contain the project proposal, and final report along with all presentations, project source code, project setup, user manual, and supporting tutorials (if applicable).
6. The submission date of hard-bound copies should be considered the completion date of the project report.
7. The final result will only be declared after receiving three hard copies of the report and CD within the specified timeline defined by the department.
8. The FYP Report should be submitted as per the schedule given below.
    a. If the FYP demo is conducted in **SPRING SEMESTER**, a student is required to submit his/ her FYP report (Hard binding) within 60 days of the last day of the final term Exam.

b. If the FYP demo is conducted in **FALL SEMESTER**, a student is required to submit his/ her FYP report (Hard binding) within 60 days of the first day of the start of the Spring semester.
c. If the student fails to fulfill the requirements (Hard binding submission), He/She will have to pay the fee equivalent to the number of credits allocated for the FYP.

3.8. Formatting of Final Report

1. Use an A4 size page with a top, bottom, and right margin of one inch and a left margin of 1.25 inches. Strictly follow margins throughout the report. No blank spaces will be left on either side.
2. Use only one side of the page for printing.
3. Times New Roman font is recommended for the whole project report.
4. The chapter title should be in 18 pt size, bold.
5. Headings/subheadings should be from 16/14 pt size to 12 pt size in bold depending upon the level of heading.
6. Body text should be in 12 pt size with line spacing 1.5
7. Body text should be justified on the both right and left sides.
8. A separator page containing the chapter (or appendix) number 18 pt size (bold) and the chapter name in 22 pt size (bold) should be placed before the start of each chapter (or appendix). This page should contain a page number.
9. The sections should be numbered with the chapter number e.g. 1.1, 1.2, and so on in the same font size and style as the section heading. The subsections should be numbered with the number of their parent sections e.g. 2.1.1, 2.1.2, and so on.
10. Only section numbers should be used/referred to in the text. No bullets or other para numbers will be used.
11. Figure and table: For caption use Times New Roman, size 10. Provide a table title at the top and a figure title below the figure. Figures and tables should be numbered with chapter numbers as a prefix, such as 2.1, 2.2, 2.3, etc.
12. Figures must be referred to in the text before they appear in the report.
13. Figures and Tables should be referred to with their number in the text.
14. References: List all the books, journals, research articles, web sites you referred to for the Project and place the list under Bibliography or References at end of your report.

The list should be numbered. Insert the number of reference materials that you learned, copied, or referred to with the text in your report. For example, a book on Java is placed at number 2 in your reference list and you are mentioning features of Java from that book in your report. You must insert [2] After writing the features of Java in your report. A general reference like Wikipedia should not be used.

15. Roman Numbering: The first few pages from the dedication to the table of contents should be separately numbered in roman numbering as (i), (ii), (iii), and so on. The normal numbering (1, 2, 3, ….) will start from the first page of chapter 1.

3.9. Sequence and Content of Final Report

1. Title Page: The title page should have the name of the project in 18 pt size (bold), a monogram of the university in 2 – 2.25” diameter, followed by the developer’s name in 16 pt size (bold). Below is the phrase Supervised by and the name of the supervisor in a similar format. The name of the university with the year of completion should be in 14 pt size (capital letters) close to the bottom of the page. The last line contains the month and year of submission.
2. Abstract: The abstract should consist of three to four paragraphs. The first paragraph will provide a project overview. Also, discuss existing systems. The next paragraph should deal with project methodology explaining what has been done and how it has been done. In the last paragraph testing, validation, and achievements should be discussed.
3. Final Approval Certificate: As per the sample given in Appendix G.
4. Declaration: As per the sample given in Appendix G.
5. Plagiarism Certificate: As per the sample given in Appendix G.
6. Turnitin Originality Certificate: As per the sample given in Appendix G.
7. Dedication (Optional): As per the sample given in Appendix G.
8. Acknowledgment (Optional): As per the sample given in Appendix G.
9. Table of Contents: The table of contents pages should not be numbered, and the contents must start from page number 1. Any page(s) before the table of contents should be numbered in Roman. The page numbers should match correctly the actual contents in the final version of the report. Heading up to the third level may be included in the table of contents as described in the sample.

10. List of Figures: All the figures used in the report are mentioned here according to their page number
11. List of Tables: All the tables used in the report are mentioned here according to their page numbers.
12. The possible sequence and organization of report chapters are given in Appendix H.
13. Appendices: Appendices should be appended at end of the project as Appendix – I, Appendix – II, and so on. There should be separate appendices for the material collected during the system study (sample forms, sample reports, etc.), extra information (conversions tables, data dictionary, definitions of terms, or any material that would help in understanding some content of the report/thesis), and user manual of the system.
14. Bibliography and References: The list of books, articles, and other sources should be listed on the last page of the report. All references must be cited in the text. All references should be written in IEEE format.

Appendices

Appendix A: Template for Project Idea

FINAL YEAR PROJECT IDEA **DEPARTMENT OF CS / SE** **NATIONAL UNIVERSITY OF MODERN LANGUAGES ISLAMABAD**	
Title	Meaningful phrases convey the idea.
Project Domain	For example web applications, mobile applications, IoT applications, etc.
Description	A short paragraph to describe the project idea.
Tools and technologies	The required tools and technologies should be mentioned to design and implement the project.
Team size	No. of students in a team required to do the project (2-3 students).
Additional information	Any other relevant information.

Appendix B: Template for Meeting Log

MEETING LOG (FYP) DEPARTMENT OF CS / SE	
SECTION 1 (to be completed by the students before the	
Title of Project	
Supervisor Name	
Student Names with roll no.	
Date	
Date of Previous Meeting	
Work done since last meeting	
Issues/tasks to be discussed	
Signature (team lead)	
SECTION 2	
Tasks assigned to students	
Date of next meeting	
Signature	

APPENDIX-C FYP-I [100 Marks]

LEARNING OUTCOMES (CLOs)

Upon successful completion of the FYP-I, the student will achieve the following:

No.	Statement of CLO	Domain	Taxonomy Level	PLO
CLO-1	**Demonstrate** a deep understanding of computing principles, pedagogical theories, and instructional design methods for addressing problems related to computing domain	Cognitive	C2	1
CLO-2	**Demonstrate** the ability to apply knowledge of computing, mathematics, and related fields in solving real-world computing problems to propose the FYP-idea by identifying all its aspects and challenges	Cognitive	C2	2
CLO-3	**Analyze** the research literature to write project proposal and work breakdown structure according to the scientific writing standards.	Cognitive	C4	3
CLO-4	**Follow** the instructions of the supervisor to devise a design or method and expected solution to get the results	Psychomotor	P3	4

CLO-5	**Present** technical information through written reports, oral presentations, and visual aids (e.g., diagrams, software interfaces) to communicate complex technical content to a non-technical audience	Affective	A2	7
CLO-6	**Demonstrate** the societal worth of the proposed solution to show how their project directly addresses a specific societal issue	Affective	A3	8
CLO-7	**Display** a professional commitment to ethical practice on a daily basis to share ideas and updates with in a team setting	Affective	A5	6
CLO-8	**Recognize** the importance of being answerable for the progress and quality of their contributions to the project	Affective	A5	9

Assessment Distribution & Mapping

Assessment	**Supervisor**	**Non-Supervisor/Committee Members**
Individual score by Supervisor	20	
Progress Presentation		40
Project Proposal		40
Total	20	80

ASSESSMENT MAPPING WITH PLO/CLO

Assessments	Evaluator	Criteria	PLO										CLO
			1	2	3	4	5	6	7	8	9	10	
Individual score by supervisor (20%)	Supervisor	Engagement in Weekly interaction						✓					7
		Tasks Progress Reporting						✓					7
		Logbook Progress Record: Understanding	✓		✓								1,3
		Logbook Progress Record: Explanation		✓	✓								2,3
		Logbook Progress Record: Supporting Reference		✓	✓								2,3
		Logbook Progress Record: Weekly Entry						✓					7
	Non supervisor/Co	Topic Knowledge (Background and Literature Review)	✓	✓	✓								1,2,3

Assessments	Evaluator	Criteria	PLO										CLO
Proposal Progress Presentation (40%)	mmittee member	Technical Approach (Problem Statement, Objective and Methodology)				✓							4
		Problem Solving Creativity (Methodology)				✓							4
		Critical Thinking (Preliminary Result/Outcome & Analysis)				✓							4
		Communication Skills							✓				5
Proposal (40%)	Non Supervisor/Committee Member	Accountability									✓		8
		Abstract							✓				5
		Background Study (Existing Systems)	✓	✓	✓								1,2,3
		Problem Statement			✓								3
		Objective and Scope/Limitation									✓		8
		Significance								✓			6
		Proposed System Comparison with Existing Systems			✓								3
		Methodology				✓							4

Assessments	Evaluator	Criteria	PLO										CLO
		Proposed System			✓								3
		Milestone/Gantt chart and Conclusion							✓				5
		References			✓								3
		Writing Skills							✓				5
		Organization							✓				5
		Quality									✓		8
		Total PLO Count in FYP-I	3	4	9	4	-	3	5	1	3		

ASSESSMENT DISTRIBUTION

Assessments	Evaluator	Criteria	Description for highest marks	Max-Marks
Individual score by supervisor (20%)	Supervisor			
		Engagement in Weekly interaction	Student actively asks questions and ENGAGES in the discussion during sessions.	5
		Tasks Progress Reporting	EXCEPTIONAL progress for the given tasks submitted on or prior to due date	3

Assessments	Evaluator	Criteria	Description for highest marks	Max-Marks
		Logbook Progress Record: Understanding	Shows EXCEPTIONAL understanding of the knowledge and concepts required to complete the project.	3
		Logbook Progress Record: Explanation	COMPREHENSIVE justification is explained for all decision made	3
		Logbook Progress Record: Supporting Reference	ALL references are relevant and recent within 5 years to support decision made	3
		Logbook Progress Record: Weekly Entry	Student have completed at least 13 weeks worth of progress update	3
		Total		**20**
Proposal Progress Presentation (40%)	Non supervisor/Committee member	Topic Knowledge (Background and Literature Review)	Displayed an EXCELLENT grasp of the project domain. Demonstrated EXCELLENT mastery of content, application and implications. EXTENSIVE research depth across range of resources through complex engineering activities.	5
		Technical Approach (Problem Statement, Objective and Methodology)	Methodology proposed FULLY supports the problem statements and objectives, while considering the consequences to society and the environment through complex engineering activities.	10
		Problem Solving Creativity (Methodology)	VERY creative and original. Imaginative design and use of resources through complex engineering activities.	10
		Critical Thinking (Preminiary Result/Outcome & Analysis)	Displays project outcome and include EXTENSIVE elaboration with HOLISTIC conclusions	10
		Communication Skills	Visual aids are OUTSTANDING, seamlessly integrated into the presentation, highly engaging, and significantly enhance understanding and retention of information.	5

Assessments	Evaluator	Criteria	Description for highest marks	Max-Marks
		Total		**40**
Proposal (40%)	Non Supervisor/Committee Member	Accountability	Proposal is submitted on or prior to due date with all submission criteria followed.	3
		Abstract	ALL aspects (background/problem/objective/methodology/results) of the report are included, and displays an EXCELLENT understanding on abstract writing.	2
		Background Study (Existing Systems)	Relevant information is compiled that COMPREHENSIVELY outlines the main topic and familiarisation of issues to form a STRONG foundation for solving complex engineering problems. The contents STRONGLY encourage readers to continue reading the report.	3
		Problem Statement	EXCEPTIONAL ability to identify and explain complex engineering problems (high level problems) and describe it in a well-structured logic.	3
		Objective and Scope/Limitation	CLEAR objectives supported with COMPREHENSIVE scope and limitations presented.	3
		Significance	Shows COMPREHENSIVE understanding and evaluation of the sustainability aspect and impact of professional engineering work to the solutions of complex engineering problems in economic, societal, and environmental contexts.	3
		Proposed System Comparison with Existing Systems	EXCEPTIONAL ability to search and evaluate relevant available literature in the relevant domain, with COMPREHENSIVE analysis on gap provided.	4

Assessments	Evaluator	Criteria	Description for highest marks	Max-Marks
		Methodology	Able to design methodology that COMPREHENSIVELY meet specified objectives with appropriate considerations to ALL of the following aspects when solving related complex engineering problems; cost, public health and safety, cultural, societal, and environmental.	4
		Proposed System	Topics were THOROUGHLY and CRITICALLY discussed in relation to existing work that underlying complex engineering problems involving many component parts or sub-problems.	5
		Milestone/Gantt chart and Conclusion	WELL SUPPORTED conclusion based on findings. Conclusions are related to objectives, and suggest significant improvement for future works	2
		References	All sources are accurately documented in the desired format.	2
		Writing Skills	Shows EXCELLENT ability to comprehend and write effective report and design documentation. ALMOST NO grammatical, spelling or punctuation errors are present.	2
		Organization	EXCELLENT organization of information based on the given format	2
		Quality	Report is EXCEPTIONAL, exhibiting depth of analysis, insightful conclusions, and outstanding organization, surpassing expectations. Demonstrates HIGH self-reliance when working independently	2
		Total		**40**

ASSESSMENT BREAKDOWN

Assessments	Evaluator	Criteria	T. Marks	Breakdown				
Individual score by supervisor (20%)	Supervisor			1 **Un Acceptable** Achievement Level <=40%.	2 **Un-Satisfactory** Achievement Level >40%-55%	3 **Developing** Achievement Level >55% - 70%.	4 **Good** Achievement Level >70%-85%.	5 **Outstanding** Achievement Level >85%.
		Engagement in Weekly interaction	5	NO participation during engagement session	Student just wait for supervisor instruction WITHOUT any initiative to ask questions	Student ask questions and WAIT for supervisor input	Student ACTIVELY asking questions	Student actively asks questions and ENGAGES in the discussion during sessions.
		Tasks Progress Reporting	3	NO progress submitted to the supervisor for the given tasks	MINIMAL progress for the given tasks but submitted on or prior to due date	SUFFICIENT progress for the given tasks is submitted on or prior to due date	COMPREHENSIVE progress for the given tasks is submitted on or prior to due date	EXCEPTIONAL progress for the given tasks submitted on or prior to due date

Assessments	Evaluator	Criteria	T. Marks	Breakdown				
		Logbook Progress Record: Understanding	3	Shows NO understanding of the knowledge and concepts required to complete the project.	Shows very LIMITED understanding of the knowledge and concepts required to complete the project.	Shows SUFFICIENT understanding of the knowledge and concepts required to complete the project.	Shows COMPREHENSIVE understanding of the knowledge and concepts required to complete the project.	Shows EXCEPTIONAL understanding of the knowledge and concepts required to complete the project.
		Logbook Progress Record: Explanation	3	NO justification is explained for all decision made	INADEQUATE justification for all decision made	SUFFICIENT justification for all decision made	THOROUGH justification is explained for all decision made	COMPREHENSIVE justification is explained for all decision made
		Logbook Progress Record: Supporting Reference	3	NO reference provided	References are NOT RELATED to support decision made	NOT ALL references are relevant and recent within 5 years to support decision made	References are relevant but NOT ALL are recent within 5 years to support decision made	ALL references are relevant and recent within 5 years to support decision made
		Logbook Progress Record: Weekly Entry	3	NO logbook submitted	Student have completed between 1-4 weeks worths of progress update	Student have completed between 5-8 weeks worths of progress update	Student have completed between 9-12 weeks worths of progress update	Student have completed at least 13 weeks worth of progress update
		Total	**20**					

Assessments	Evaluator	Criteria	T. Marks	Breakdown				
Proposal Progress Presentation (40%)	Non supervisor/ Committee member	Topic Knowledge (Background and Literature Review)	5	NO participation in progress Presentation	Displayed a POOR grasp of the project domain. Demonstrated a SUPERFICIAL handling of content, application and implications. LITTLE DEPTH across range of resources through complex engineering activities.	Displayed ADEQUATE grasp of the project domain. Demonstrated ADEQUATE mastery of content, application and implications. SUFFICIENT depth across range of resources through complex engineering activities.	Displayed STRONG grasp of the project domain. Demonstrated GOOD mastery of content, application and implications. SUBSTANTIAL depth across range of resources through complex engineering activities.	Displayed an EXCELLENT grasp of the project domain. Demonstrated EXCELLENT mastery of content, application and implications. EXTENSIVE depth across range of resources through complex engineering activities.
		Technical Approach (Problem Statement, Objective and Methodology)	10	NO participation in Progress Presentation	Methodology presented DOES NOT support the problem statements and objectives, the consequences to society and the environment through complex engineering activities.	Methodology presented ADEQUATELY supports the problem statements and objectives, while considering the consequences to society and the environment through complex engineering activities.	Methodology presented STRONGLY supports the problem statements and objectives, while considering the consequences to society and the environment through complex engineering activities.	Methodology proposed FULLY supports the problem statements and objectives, while considering the consequences to society and the environment through complex engineering activities.

Assessments	Evaluator	Criteria	T. Marks	Breakdown				
		Problem Solving Creativity (Methodology)	10	NO participation in progress Presentation	LACKED creativity. Very ordinary and mundane through complex engineering activities.	Routine treatment, MINIMAL thought given to originality or creativity through complex engineering activities.	Routine treatment, SUFFICIENT thought given to originality or creativity through complex engineering activities.	VERY creative and original. Imaginative design and use of resources through complex engineering activities.
		Critical Thinking (Preminiary Result/Outcome & Analysis)	10	NO participation in Progress Presentation	Displays some project outcome but NOT ABLE to elaborate or explain the presented outcome	Displays project outcome and include ADEQUATE elaboration and explanation to the presented outcome	Displays project outcome and include COMPREHENSIVE elaboration and explanation to the presented outcome	Displays project outcome and include EXTENSIVE elaboration with HOLISTIC conclusions
		Communication Skills	5	NO participation in progress Presentation	DIFFICULT to hear, occasional eye contact, some mumbling, little or no expression, nervous, some distracting mannerisms, reads much of slide.	Clear voice, but LACKING expressiveness, slight nervousness, and less polished delivery.	Clear, EXPRESSIVE voice, poised demeanor, maintains good posture, and exhibits no distracting mannerisms.	Visual aids are OUTSTANDING, seamlessly integrated into the presentation, highly engaging, and significantly enhance understanding and retention of information.
		Total	**40**					

Assessments	Evaluator	Criteria	T. Marks	Breakdown				
Proposal (40%)	Non Supervisor/ Committee Member	Accountability	3	NO proposal submitted.	Proposal is submitted after week 15	Proposal is submitted after due date	Proposal is submitted on or prior to due date but criteria for submission are not followed.	Proposal is submitted on or prior to due date with all submission criteria followed.
		Abstract	2	NO abstract presented	SEVERAL aspects (background/problem/objective/methodology/results) of the report are missing, and displays a LACK OF UNDERSTANDING on abstract writing.	MOST aspects (background/problem/objective/methodology/results) of the report are included, and displays SUFFICIENT understanding on abstract writing.	MOST aspects (background/problem/objective/methodology/results) of the report are included, and displays a GOOD understanding on abstract writing.	ALL aspects (background/problem/objective/methodology/results) of the report are included, and displays an EXCELLENT understanding on abstract writing.

Assessments	Evaluator	Criteria	T. Marks	Breakdown				
		Background Study (Existing Systems)	3	NO background study presented	Information compiled is IRRELEVANT to the main topic and familiarisation of issues for the study, resulting in LITTLE TO NO foundation for solving complex engineering problems.	Relevant information is compiled that SUFFICIENTLY outlines the main topic and familiarisation of issues to form an ADEQUATE foundation for solving complex engineering problems.	Relevant information is compiled that provides a GOOD outline to the main topic and familiarisation of issues to form a GOOD foundation for solving complex engineering problems. The contents encourage readers to continue reading the report.	Relevant information is compiled that COMPREHENSIVELY outlines the main topic and familiarisation of issues to form a STRONG foundation for solving complex engineering problems. The contents STRONGLY encourage readers to continue reading the report.
		Problem Statement	3	NO problem statement presented	UNABLE to identify and explain the specific complex engineering problems (high level problems) that is going to be solved in the research.	SUFFICIENT ability to identify and explain complex engineering problems (high level problems) but the construction of logic is not really well structured	GOOD ability to identify and explain complex engineering problems (high level problems) but the construction of logic is not really well structured.	EXCEPTIONAL ability to identify and explain complex engineering problems (high level problems) and describe it in a well-structured logic.

Assessments	Evaluator	Criteria	T. Marks	Breakdown				
		Objective and Scope/Limitation	3	NO objective and NO scope/limitation presented	UNCLEAR objectives but NO scope/limitations presented.	CLEAR objectives but NO scope/limitation presented	CLEAR objectives supported with SOME scope/limitations presented.	CLEAR objectives supported with COMPREHENSIVE scope and limitations presented.
		Significance	3	NO research significance presented	UNABLE to understand and evaluate the sustainability aspect and the impact of professional engineering work on the solutions to complex engineering problems in economic, societal, and environmental contexts.	Shows FAIR understanding and evaluation of the sustainability aspect and impact of professional engineering work to the solutions of complex engineering problems in economic, societal and environmental contexts.	Shows GOOD understanding and evaluation of the sustainability aspect and impact of professional engineering work to the solutions of complex engineering problems in economic, societal and environmental contexts.	Shows COMPREHENSIVE understanding and evaluation of the sustainability aspect and impact of professional engineering work to the solutions of complex engineering problems in economic, societal, and environmental contexts.

Assessments	Evaluator	Criteria	T. Marks	Breakdown				
		Proposed System Comparison with Existing Systems	4	NO literature review presented	UNABLE to search and evaluate relevant available literature in the relevant domain.	Shows FAIR ablity to search and evaluate relevant available literature in the relevant domain without any gaps presented, or gap presented but WITHOUT meaningful analysis.	Shows GOOD ability to search and evaluate relevant available literature in the relevant domain, with SUFFICIENT analysis on gap provided.	EXCEPTIONAL ability to search and evaluate relevant available literature in the relevant domain, with COMPREHENSIVE analysis on gap provided.
		Methodology	4	NO methodology presented	UNABLE to design methodology that meets specified objectives with appropriate consideration for cost, public health and safety, cultural, societal, and environmental aspects to solve related complex engineering problems.	Able to design methodology that ADEQUATELY meet specified objectives but with LITTLE TO NO considerations on the following aspects when solving related comlex engineering problems; cost, public health and safety, cultural, societal, and environmental.	Able to design methodology that ADEQUATELY meet specified objectives with appropriate considerations to SOME of the following aspects when solving related comlex engineering problems; cost, public health and safety, cultural, societal, and environmental.	Able to design methodology that COMPREHENSIVELY meet specified objectives with appropriate considerations to ALL of the following aspects when solving related complex engineering problems; cost, public health and safety, cultural, societal, and environmental.

Assessments	Evaluator	Criteria	T. Marks	Breakdown				
		Proposed System	5	NO preliminary results or critical discussion presented	INCORRECT or few links made with underlying complex engineering problems involving many component parts or sub-problems.	SUFFICIENT sections discuss the underlying complex engineering problems with little critical integration involving many component parts or sub-problems.	SUBSTANTIAL sections discuss the underlying complex engineering problems with little critical integration involving many component parts or sub-problems.	Topics were THOROUGHLY and CRITICALLY discussed in relation to existing work that underlying complex engineering problems involving many component parts or sub-problems.
		Milestone /Gantt chart and Conclusion	2	NO conclusion and future work presented	WEAK conclusion (not directly related to objective), and no suggestion for future works.	Conclusion PARTIALLY supported by results/findings but no suggestion of future works.	Conclusion PARTIALLY supported by results/findings, and suggest future works.	WELL SUPPORTED conclusion based on findings. Conclusions are related to objectives, and suggest significant improvement for future works
		References	2	NO references presented.	More than 50% sources are not accurately documented.	More than 50% and less than 70% references are accurately documented in the desired format.	More than 70% and less than 100% references are accurately documented in the desired format.	All sources are accurately documented in the desired format.

Assessments	Evaluator	Criteria	T. Marks	Breakdown				
		Writing Skills	2	NO report submitted.	NOT ABLE to comprehend and write effective report and design documentation. MANY grammatical, spelling or punctuation errors are present.	Shows MODERATE ability to comprehend and write effective report and design documentation. A FEW grammatical, spelling, or punctuation errors are present.	Shows SUFFICIENT ability to comprehend and write effective report and design documentation. MINIMAL grammatical, spelling or punctuation errors are present.	Shows EXCELLENT ability to comprehend and write effective report and design documentation. ALMOST NO grammatical, spelling or punctuation errors are present.
		Organization	2	NO report submitted.	POOR organization of information based on the given format	MODERATE organization of information based on the given format	GOOD organization of information based on the given format	EXCELLENT organization of information based on the given format
		Quality	2	NO report submitted.	Report is INCOMPLETE, contains significant errors, or LACKS coherence. Has a LITTLE self-reliance when working independently	Report is ADEQUATE, with sufficient detail, organization, or analysis. Has FAIR self-reliance when working independently	Report is WELL-DEVELOPED, demonstrating a clear understanding of the topic, thorough analysis, and effective organization. Demonstrates a GOOD level of self-reliance when working independently.	Report is EXCEPTIONAL, exhibiting depth of analysis, insightful conclusions, and outstanding organization, surpassing expectations. Demonstrates HIGH self-reliance when working independently

Assessments	Evaluator	Criteria	T. Marks	Breakdown				
		Total	40					

FYP-II [100 Marks]

COURSE LEARNING OUTCOMES (CLOs):

Upon successful completion of the course, the student will achieve the following:

No.	Statement of CLO	Domain	Taxonomy Level	PLO
CLO-1	**Apply** the computing knowledge to use appropriate techniques, resources, and modern engineering tools with an understanding of their limitations to complete FYP	Psychomotor	P4	5
CLO-2	**Construct** and document the architecture and design of the project to integrate various system components, technologies, and to ensure alignment with both functional and non-functional requirements	Psychomotor	P5	4
CLO-3	**Write** functional code of the project to create working modules of the project	Psychomotor	P5	4

CLO-4	**Evaluate** the quality and significance of the results, comparing them to initial goals, literature benchmarks, or expectations, and deciding whether the project outcomes are successful or require improvement	Cognitive	C5	4
CLO-5	**Formulate** a formal final report and present the project in professional and effective way	Affective	A4	7
CLO-6	**Organize** and integrate different viewpoints during the defense, presenting a balanced perspective	Affective	A4	7
CLO-7	Consistently **apply** ethical principles across the project lifecycle, from ideation to deployment, ensuring transparency in reporting project limitations and potential biases	Affective	A5	9
CLO-8	**Explain** how the individual and collaborative work of students contributed to the overall project goals and how their contributions fit into the larger system	Cognitive	C2	6
CLO-9	**Evaluate** the outcomes of the project, assessing the effectiveness of different approaches and making recommendations for improvement based on evidence	Cognitive	C5	10

Assessment Distribution & Mapping

Assessment	**Supervisor**	**Non-Supervisor/Committee Members**
Individual score by Supervisor	20	

Assessment	**Supervisor**	**Non-Supervisor/Committee Members**
Project Presentation		40
Project Report/Thesis		40
Total	20	80

ASSESSMENT MAPPING WITH PLO/CLO

Assessments	**Evaluator**	**Criteria**	**PLO**										**CLO**
			1	2	3	4	5	6	7	8	9	10	
Individual score by supervisor (20%)	Supervisor	Engagement in Weekly interaction									✓		7
		Tasks Progress Reporting									✓		7

Assessments	Evaluator	Criteria	PLO										CLO
		Logbook Progress Record: Understanding									✓		7
		Logbook Progress Record: Explanation									✓		7
		Logbook Progress Record: Supporting Reference									✓		7
		Logbook Progress Record: Weekly Entry									✓		7
Project Presentation/Demo (40%)	Non supervisor/Committee member	Topic Knowledge (Background and Literature Review)					✓		✓				1,6
		Technical Approach (Problem Statement, Objective and Methodology)				✓	✓		✓				1,2,6
		Problem Solving Creativity (Methodology)				✓	✓		✓				1,2,6
		Critical Thinking (Result/Outcome & Analysis)				✓	✓		✓				3,4,6
		Communication Skills							✓				5,6
Project Report (40%)	Non Supervisor/Committee Member	Accountability						✓					8
		Abstract							✓				5
		Chapter 1: Background Study							✓				5

Assessments	Evaluator	Criteria	PLO										CLO
		Chapter 1: Problem Statement							✓				5
		Chapter 1: Objective and Scope/Limitation							✓				5
		Chapter 1: Significance							✓				5
		Chapter 2: Background and Existing work							✓				5
		Chapter 3: System Model/Proposed System				✓							2
		Chapter 4: Hardware/ Software Design				✓							2
		Chapter 4 Algorithm/Source Code				✓							3
		Chapter 5: Testing Results & discussion				✓						✓	4,9
		Chapter 6: Conclusion & Future works							✓				5
		References							✓				5
		Writing Skills							✓				5
		Organization							✓				5

Assessments	Evaluator	Criteria	PLO										CLO
		Quality										✓	9
		Total PLO Count in FYP-II				7	4	1	15		6	2	

<u>ASSESSMENT DISTRIBUTION</u>

Assessments	Evaluator	Criteria	Description for highest marks	Max-Marks
Individual score by supervisor (20%)	Supervisor			
		Engagement in Weekly interaction	Student actively asking questions and always gives suggestions/recommedations to problems faced in the project	5
		Tasks Progress Reporting	Comprehensive progress for the given tasks submitted on or prior to due date	3
		Logbook Progress Record: Understanding	Shows sufficient understanding of the knowledge and concepts required to complete the project.	3
		Logbook Progress Record: Explanation	The explanation provided clear reasoning for all decision made	3

Assessments	Evaluator	Criteria	Description for highest marks	Max-Marks
		Logbook Progress Record: Supporting Reference	All references are relevant and updated to support decision made	3
		Logbook Progress Record: Weekly Entry	Student have completed at least 13 weeks worth of progress update	3
		Total		**20**
Project Presentation (40%)	Non supervisor/Committee member	Topic Knowledge (Background and Literature Review)	Displayed an excellent grasp of the project domain. Demonstrated excellent knowledge of content, application and implications. Excellent research depth exhibited across a range of resources when executing complex engineering activities.	5
		Technical Approach (Problem Statement, Objective and Methodology)	Methodology proposed strongly supports the problem statements and objectives, while considering the consequences to society and the environment when executing complex engineering activities.	10
		Problem Solving Creativity (Methodology)	Very creative and original. Imaginative design and use of materials through complex engineering activities.	10
		Critical Thinking (Expected outcomes/Results & Analysis)	Displays project outcome and include extensive elaboration and critical explanation to the presented outcome	10
		Communication Skills	Visual aids are outstanding, seamlessly integrated into the presentation, highly engaging, and significantly enhance understanding and retention of information.	5

Assessments	Evaluator	Criteria	Description for highest marks	Max-Marks
		Total		**40**
Project Report (40%)	Non Supervisor/Committee Member	Accountability	Report is submitted on or prior to due date with all submission criteria followed.	3
		Abstract	ALL aspects (background/problem/objective/methodology/results) of the report are included, and displays an EXCELLENT understanding on abstract writing.	1
		Chapter 1: Background Study	Relevant information is compiled that COMPREHENSIVELY outlines the main topic and familiarisation of issues to form a STRONG foundation for solving complex engineering problems. The contents STRONGLY encourage readers to continue reading the report.	2
		Chapter 1: Problem Statement	EXCEPTIONAL ability to identify and explain complex engineering problems (high level problems) and describe it in a well-structured logic.	2
		Chapter 1: Objective and Scope/Limitation	CLEAR objectives supported with COMPREHENSIVE scope and limitations presented.	2
		Chapter 1: Significance	Shows COMPREHENSIVE understanding and evaluation of the sustainability aspect and impact of professional engineering work to the solutions of complex engineering problems in economic, societal, and environmental contexts.	2
		Chapter 2: Background and Existing work	EXCEPTIONAL ability to search and evaluate relevant available literature in the relevant domain, with COMPREHENSIVE analysis on gap provided.	4

Assessments	**Evaluator**	**Criteria**	**Description for highest marks**	**Max-Marks**
		Chapter 3: System Model/Proposed System	Able to design methodology that COMPREHENSIVELY meet specified objectives with appropriate considerations to ALL of the following aspects when solving related comlex engineering problems; cost, public health and safety, cultural, societal, and environmental.	5
		Chapter 4: Hardware/Software Design	Hardware/software works as intended with full functionality. The system meets all specified requirements and runs without major issues.	5
		Chapter 4: Algorithm/Source Code	Code is well-organized, with clear and consistent structure. It follows best practices such as modularization, clear naming conventions, and appropriate use of comments	5
		Chapter 5: Testing Results & discussion	Results are presented clearly and accurately, with appropriate use of graphs, tables, or charts. The data is well-organized and easy to understand.	2
		Chapter 6: Conclusion & Future works	EXCELLENT conclusion that relates findings/results with the objectives. Suggestions for future work justified with COMPREHENSIVE considerations of the limitations in the current study.	2
		References	ALL cited sources in the Reference section and throughout the thesis are accurately documented in the given format. 50% references are from literature that have been published in the LAST 5 years.	1
		Writing Skills	Shows EXCELLENT ability to comprehend and write effective report and design documentation. ALMOST NO grammatical, spelling or punctuation errors are present.	1
		Organization	EXCELLENT organization of information based on the given format	1
		Quality	Report is EXCEPTIONAL, exhibiting depth of analysis, insightful conclusions, and outstanding organization, surpassing expectations. Demonstrates HIGH self-reliance when working independently	2

Assessments	Evaluator	Criteria	Description for highest marks	Max-Marks
		Total		**40**

ASSESSMENT BREAKDOWN

Assessments	Evaluator	Criteria	T. Marks	Breakdown				
Individual score by supervisor (20%)	Supervisor			1 **Un Acceptable** Achievement Level <=40%.	2 **Un-Satisfactory** Achievement Level >40%-55%	3 **Developing** Achievement Level >55% - 70%.	4 **Good** Achievement Level >70%-85%.	5 **Outstanding** Achievement Level >85%.
		Engagement in Weekly interaction	5	NO participation during engagement session	Student just wait for supervisor instruction WITHOUT any initiative to ask questions	Student ask questions and WAIT for supervisor input	Student ACTIVELY asking questions	Student actively asking questions and always gives suggestions/recommedations to problems faced in the project

Assessments	Evaluator	Criteria	T. Marks	Breakdown				
		Tasks Progress Reporting	3	NO progress submitted to the supervisor for the given tasks	MINIMAL progress for the given tasks but submitted on or prior to due date	SUFFICIENT progress for the given tasks is submitted on or prior to due date	COMPREHENSIVE progress for the given tasks is submitted on or prior to due date	Comprehensive progress for the given tasks submitted on or prior to due date
		Logbook Progress Record: Understanding	3	Shows NO understanding of the knowledge and concepts required to complete the project.	Shows very LIMITED understanding of the knowledge and concepts required to complete the project.	Shows SUFFICIENT understanding of the knowledge and concepts required to complete the project.	Shows COMPREHENSIVE understanding of the knowledge and concepts required to complete the project.	Shows sufficient understanding of the knowledge and concepts required to complete the project.
		Logbook Progress Record: Explanation	3	NO justification is explained for all decision made	INADEQUATE justification for all decision made	SUFFICIENT justification for all decision made	THOROUGH justification is explained for all decision made	The explanation provided clear reasoning for all decision made
		Logbook Progress Record: Supporting Reference	3	NO reference provided	References are NOT RELATED to support decision made	NOT ALL references are relevant and recent within 5 years to support decision made	References are relevant but NOT ALL are recent within 5 years to support decision made	All references are relevant and updated to support decision made

Assessments	Evaluator	Criteria	T. Marks	Breakdown				
		Logbook Progress Record: Weekly Entry	3	NO logbook submitted	Student have completed between 1-4 weeks worths of progress update	Student have completed between 5-8 weeks worths of progress update	Student have completed between 9-12 weeks worths of progress update	Student have completed at least 13 weeks worth of progress update
		Total	**20**					
Project Presentation/Demo (40%)	Non supervisor/ Committee member	Topic Knowledge (Background and Literature Review)	5	NO participation in project Presentation/demo	Displayed a POOR grasp of the project domain. Demonstrated a SUPERFICIAL handling of content, application and implications. LITTLE DEPTH across range of resources through complex engineering activities.	Displayed ADEQUATE grasp of the project domain. Demonstrated ADEQUATE mastery of content, application and implications. SUFFICIENT depth across range of resources through complex engineering activities.	Displayed STRONG grasp of the project domain. Demonstrated GOOD mastery of content, application and implications. SUBSTANTIAL depth across range of resources through complex engineering activities.	Displayed an excellent grasp of the project domain. Demonstrated excellent knowledge of content, application and implications. Excellent research depth exhibited across a range of resources when executing complex engineering activities.

Assessments	Evaluator	Criteria	T. Marks	Breakdown				
		Technical Approach (Problem Statement, Objective and Methodology)	10	NO participation in project Presentation/demo	Methodology presented DOES NOT support the problem statements and objectives, the consequences to society and the environment through complex engineering activities.	Methodology presented ADEQUATELY supports the problem statements and objectives, while considering the consequences to society and the environment through complex engineering activities.	Methodology presented STRONGLY supports the problem statements and objectives, while considering the consequences to society and the environment through complex engineering activities.	Methodology proposed strongly supports the problem statements and objectives, while considering the consequences to society and the environment when executing complex engineering activities.
		Problem Solving Creativity (Methodology)	10	NO participation in project Presentation/demo	LACKED creativity. Very ordinary and mundane through complex engineering activities.	Routine treatment, MINIMAL thought given to originality or creativity through complex engineering activities.	Routine treatment, SUFFICIENT thought given to originality or creativity through complex engineering activities.	Very creative and original. Imaginative design and use of materials through complex engineering activities.
		Critical Thinking (Expected outcomes /Results & Analysis)	10	NO participation in project Presentation/demo	Displays some project outcome but NOT ABLE to elaborate or explain the presented outcome	Displays project outcome and include ADEQUATE elaboration and explanation to the presented outcome	Displays project outcome and include COMPREHENSIVE elaboration and explanation to the presented outcome	Displays project outcome and include extensive elaboration and critical explanation to the presented outcome

Assessments	Evaluator	Criteria	T. Marks	Breakdown				
		Communication Skills	5	NO participation in project Presentation/demo	DIFFICULT to hear, occasional eye contact, some mumbling, little or no expression, nervous, some distracting mannerisms, reads much of slide.	Clear voice, but LACKING expressiveness, slight nervousness, and less polished delivery.	Clear, EXPRESSIVE voice, poised demeanor, maintains good posture, and exhibits no distracting mannerisms.	Visual aids are outstanding, seamlessly integrated into the presentation, highly engaging, and significantly enhance understanding and retention of information.
		Total	**40**					
Project Report (40%)	Non Supervisor/ Committee Member	Accountability	3	NO report submitted.	Report is submitted after week 15	Report is submitted after due date	Report is submitted on or prior to due date but criteria for submission are not followed.	Report is submitted on or prior to due date with all submission criteria followed.
		Abstract	1	NO abstract presented	SEVERAL aspects (background/problem/objective/methodology/results) of the report are missing, and displays a LACK OF UNDERSTANDING on abstract writing.	MOST aspects (background/problem/objective/methodology/results) of the report are included, and displays SUFFICIENT understanding on abstract writing.	MOST aspects (background/problem/objective/methodology/results) of the report are included, and displays a GOOD understanding on abstract writing.	ALL aspects (background/problem/objective/methodology/results) of the report are included, and displays an EXCELLENT understanding on abstract writing.

Assessments	Evaluator	Criteria	T. Marks	Breakdown				
		Chapter 1: Background Study	2	NO background study presented	Information compiled is IRRELEVANT to the main topic and familiarisation of issues for the study, resulting in LITTLE TO NO foundation for solving complex engineering problems.	Relevant information is compiled that SUFFICIENTLY outlines the main topic and familiarisation of issues to form an ADEQUATE foundation for solving complex engineering problems.	Relevant information is compiled that provides a GOOD outline to the main topic and familiarisation of issues to form a GOOD foundation for solving complex engineering problems. The contents encourage readers to continue reading the report.	Relevant information is compiled that COMPREHENSIVELY outlines the main topic and familiarisation of issues to form a STRONG foundation for solving complex engineering problems. The contents STRONGLY encourage readers to continue reading the report.
		Chapter 1: Problem Statement	2	NO problem statement presented	UNABLE to identify and explain the specific complex engineering problems (high level problems) that is going to be solved in the research.	SUFFICIENT ability to identify and explain complex engineering problems (high level problems) but the construction of logic is not really well structured	GOOD ability to identify and explain complex engineering problems (high level problems) but the construction of logic is not really well structured.	EXCEPTIONAL ability to identify and explain complex engineering problems (high level problems) and describe it in a well-structured logic.

Assessments	Evaluator	Criteria	T. Marks	Breakdown				
		Chapter 1: Objective and Scope/Limitation	2	NO objective and NO scope/limitation presented	UNCLEAR objectives but NO scope/limitations presented.	CLEAR objectives but NO scope/limitation presented	CLEAR objectives supported with SOME scope/limitations presented.	CLEAR objectives supported with COMPREHENSIVE scope and limitations presented.
		Chapter 1: Significance	2	NO work significance presented	UNABLE to understand and evaluate the sustainability aspect and the impact of professional engineering work on the solutions to complex engineering problems in economic, societal, and environmental contexts.	Shows FAIR understanding and evaluation of the sustainability aspect and impact of professional engineering work to the solutions of complex engineering problems in economic, societal and environmental contexts.	Shows GOOD understanding and evaluation of the sustainability aspect and impact of professional engineering work to the solutions of complex engineering problems in economic, societal and environmental contexts.	Shows COMPREHENSIVE understanding and evaluation of the sustainability aspect and impact of professional engineering work to the solutions of complex engineering problems in economic, societal, and environmental contexts.

Assessments	Evaluator	Criteria	T. Marks	Breakdown				
		Chapter 2: Background and Existing work	4	NO literature review presented	UNABLE to search and evaluate relevant available literature in the relevant domain.	Shows FAIR ablity to search and evaluate relevant available literature in the relevant domain without any gaps presented, or gap presented but WITHOUT meaningful analysis.	Shows GOOD ability to search and evaluate relevant available literature in the relevant domain, with SUFFICIENT analysis on gap provided.	EXCEPTIONAL ability to search and evaluate relevant available literature in the relevant domain, with COMPREHENSIVE analysis on gap provided.
		Chapter 3: System Model/Proposed System	5	NO methodology presented	UNABLE to design methodology that meets specified objectives with appropriate consideration for cost, public health and safety, cultural, societal, and environmental aspects to solve related complex engineering problems.	Able to design methodology that ADEQUATELY meet specified objectives but with LITTLE TO NO considerations on the following aspects when solving related comlex engineering problems; cost, public health and safety, cultural, societal, and environmental.	Able to design methodology that ADEQUATELY meet specified objectives with appropriate considerations to SOME of the following aspects when solving related comlex engineering problems; cost, public health and safety, cultural, societal, and environmental.	Able to design methodology that COMPREHENSIVELY meet specified objectives with appropriate considerations to ALL of the following aspects when solving related comlex engineering problems; cost, public health and safety, cultural, societal, and environmental.

Assessments	Evaluator	Criteria	T. Marks	Breakdown				
		Chapter 4: Hardware /Software Design	5	NO design presented	Implementation is incomplete, with substantial functionality missing or failing to work as required.	System is partially functional, but key features or requirements are missing or have significant issues.	System functions well with minor issues. Most requirements are met, though there may be slight deviations or bugs.	Hardware/software works as intended with full functionality. The system meets all specified requirements and runs without major issues.
		Chapter 4: Algorithm /Source Code	5	Incomplete code presented	Code is unstructured, difficult to read, and lacks any meaningful comments or organization.	Code is somewhat disorganized, with unclear naming conventions and minimal commenting, making it difficult to follow.	Code is structured logically, but may have minor issues with organization, such as unclear variable names or lack of sufficient comments.	Code is well-organized, with clear and consistent structure. It follows best practices such as modularization, clear naming conventions, and appropriate use of comments

Assessments	Evaluator	Criteria	T. Marks	Breakdown				
		Chapter 5: Testing Results & discussion	2	No results presented	Results are unclear, incomplete, or inaccurate, making it difficult to evaluate the effectiveness of the solution.	Results are presented, but they may lack clarity, organization, or depth. There are occasional inaccuracies or gaps in the data provided.	Results are generally accurate and well-organized but could be presented more clearly or with additional visualization. Some minor inconsistencies may exist.	Results are presented clearly and accurately, with appropriate use of graphs, tables, or charts. The data is well-organized and easy to understand.
		Chapter 6: Conclusion & Future works	2	NO conclusion and future work presented	WEAK conclusion (not directly related to objective), and no suggestion for future works.	Conclusion PARTIALLY supported by results/findings but no suggestion of future works.	Conclusion PARTIALLY supported by results/findings, and suggest future works.	EXCELLENT conclusion that relates findings/results with the objectives. Suggestions for future work justified with COMPREHENSIVE considerations of the limitations in the current study.

Assessments	Evaluator	Criteria	T. Marks	Breakdown				
		References	1	NO references presented.	More than 50% sources are not accurately documented.	More than 50% and less than 70% references are accurately documented in the desired format.	More than 70% and less than 100% references are accurately documented in the desired format.	ALL cited sources in the Reference section and throughout the thesis are accurately documented in the given format. 50% references are from literature that have been published in the LAST 5 years.
		Writing Skills	1	NO report submitted.	NOT ABLE to comprehend and write effective report and design documentation. MANY grammatical, spelling or punctuation errors are present.	Shows MODERATE ability to comprehend and write effective report and design documentation. A FEW grammatical, spelling, or punctuation errors are present.	Shows SUFFICIENT ability to comprehend and write effective report and design documentation. MINIMAL grammatical, spelling or punctuation errors are present.	Shows EXCELLENT ability to comprehend and write effective report and design documentation. ALMOST NO grammatical, spelling or punctuation errors are present.
		Organization	1	NO report submitted.	POOR organization of information based on the given format	MODERATE organization of information based on the given format	GOOD organization of information based on the given format	EXCELLENT organization of information based on the given format

Assessments	Evaluator	Criteria	T. Marks	Breakdown				
		Quality	2	NO report submitted.	Report is INCOMPLETE, contains significant errors, or LACKS coherence. Has a LITTLE self-reliance when working independently	Report is ADEQUATE, with sufficient detail, organization, or analysis. Has FAIR self-reliance when working independently	Report is WELL-DEVELOPED, demonstrating a clear understanding of the topic, thorough analysis, and effective organization. Demonstates a GOOD level of self-reliance when working independently.	Report is EXCEPTIONAL, exhibiting depth of analysis, insightful conclusions, and outstanding organization, surpassing expectations. Demonstrates HIGH self-reliance when working independently
		Total	**40**					

Appendix D: Template for Final Report

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PROJECT NAME

Student Name 1

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Do not use Ms/Mr/Sir , can use Dr./Prof./Engr.

Student Name 2

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Student Name 3

Student Name 4

Supervised By

Supervisor Name

Submitted for the partial fulfillment of BS Computer Science / Software Engineering /BS Information Technology/ BS Artificial Intelligence degree to the Faculty of Engineering & CS

NATIONAL UNIVERSITY OF MODERN LANGUAGES

ISLAMABAD

Month, Year

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ABSTRACT

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The abstract should consist of three to four paragraphs. The first paragraph will provide a project overview. Also, discuss existing systems.

The next paragraph should deal with project methodology explaining what has been done and how it has been done.

In the last paragraph testing, validation, and achievements should be discussed.

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CERTIFICATE

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Dated: ________________

Final Approval

It is certified that the project report titled '**Project Name**' submitted by **Student name 1**, **Student name 2,** and **student name 3** for the partial fulfillment of the requirement of **"Bachelors Degree in Software Engineering/Computer Science/ Information Technology/ Artificial Intelligence"** is approved.

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COMMITTEE

Dr. FirstName LastName
DEAN FEC Signature: ________________________

Dr. Sumaria Nazir
HoD Engineering Signature: ________________________

FirstName LastName
Head Project Committee Signature: ________________________

FirstName LastName
Supervisor Signature: ________________________

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DECLARATION

We hereby declare that our dissertation is entirely our work and genuine/original. We understand that in case of discovery of any PLAGIARISM at any stage, our group will be assigned an F (FAIL) grade and it may result in withdrawal of our Bachelor's degree.

Group Members	**Signature**
1. Student Name 1	____________________
2. Student Name 2	____________________
3. Student Name 3	____________________

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PLAIGRISM CERTIFICATE

This is to certify that the project entitled "**Project Title**", is being submitted here for the award of the "**Degree of Bachelor" in "Software Engineering/Computer Science/ Information Technology/Artificial Intelligence**". This is the result of the original work by **Student Name 1, Student Name 2,** and **Student Name 3** under my supervision and guidance. The work embodied in this project has not been done earlier for the basis of the award of any degree or compatible certificate or similar title of this for any other diploma/examining body or university to the best of my knowledge and belief.

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ACKNOWLEDGMENT

(Optional)

Students may acknowledge the persons who supported them in the project work but this should be very brief and precise.

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TABLE OF CONTENTS

Chapter

Page

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LIST OF FIGURES

Figure	Caption	Page

Captions should be exactly same as in text

Screen shots and photographs should be avoided

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LIST OF TABLES

Table **Caption** **Page**

Captions should be the same as in the text

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CHAPTER NUMBER (e.g. 1)

CHAPTER TITLE (e.g. Introduction)

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APPENDICES

Appendices should be appended at end of the project as Appendix – I, Appendix – II, and so on. There should be separate appendices for the material collected during the system study (sample forms, sample reports, etc.), extra information (conversions tables, data dictionary, definitions of terms, or any material that would help in understanding some content of the report/thesis), and user manual of the system.

REFERENCES

The list of books, articles, and other sources should be listed on the last page of the report. All references must be used/cited in the text. All references should be written in IEEE format. The general format is as follows:

Book

1. W.K. Chen. Linear Networks and Systems. Belmont, CA: Wadsworth, 1993, pp. 123-35.

Book Chapters

2. J.E. Bourne. "Synthetic structureof industrial plastics," in Plastics, 2nd ed., vol. 3. J.Peters, Ed. New York:McGraw-Hill, 1964,pp. 15-67.

Article in a Journal

3. G. Pevere. "Infrared Nation." The International Journal of Infrared Design, vol. 33, pp. 56-99, Jan. 1979.

Articles from Conference Proceedings (Published)

4. D.B. Payne and H.G. Gunhold. "Digital Sundials ang broadband technology," in Proc. IOOC-ECOC, 1986,PP. 557-998.

Papers Presented at Conferences (Published)

5. B. Brandli and M. Dick. "Engineering names and concepts," presented at the 2nd Int. Conf. Engineering Education, Frankfurt, Germany, 1999.

Note: For details refer to IEEE Citation Style Guide

Appendix D: Possible Organization of the Final Report

Chapter 1	**Introduction**
1.1	Introduction (You can add before motivation some other headings necessary to support your onward headings)
1.2	Motivation
1.3	Problem statement
1.4	Goals and Objectives
1.5	5 Scope of the study
1.6	Process model (Choice of model for your project and why you have chosen)
1.7	Nature of the project (Like the web, android, IOT, AI, Machine learning, etc. give some meaningful heading)
1.8	Overview/Organization of the report
Chapter 2	**Background and Existing Work**
2.1	Introduction
2.2	Explanation of important constructs of the application domain (Explain the domain knowledge with different headings)
2.3	Existing studies/systems
2.4	Comparison of existing systems
2.5	Summary
Chapter 3	**Requirements Specification**
3.1	Introduction
3.2	Interface Requirements (Interface requirements state the mandatory things that we need to have to interface the different components of the system with themselves to make these communicate easily and compile the whole system. In this section, we also need to know what we need to have to get the system communicating with the other environment. Since project contains hardware and software both and including an online interface as well as mobile application platform. Therefore, the section should be divided into the respective categories.) 3.2.1 Hardware Interface Requirements 3.2.2 Software Interface Requirements
3.3	Functional requirements
3.4	Use case model (along with diagram)
3.5	Use cases (Use case description) 3.5.1 Use case 1 3.5.n Use case n
3.6	Non-functional requirements (which are necessary for your system)

	3.6.1 Performance 3.6.2 Reliability 3.6.3 Security 3.6.4 Consistency etc.
3.7	Resource requirements (Resources for a project include:) Equipment (H/W & S/W tools & technologies as per your project with justification/reason) Funds (Optional) Human effort (Task division/breakdown & Man months)
3.8	Database Requirements
3.9	Project Feasibility (only write which are applicable to your project) Feasibility study of the project is performed to analyze whether the project is feasible within the time and budget. 3.9.1 Technical Feasibility (e.g. Technically this project is feasible as it provides desired features via using all its components and processing their data. It does not require a very high machine to run on. It also covers all the aspects of usage, i.e. desktop application, mobile application.) 3.9.2 Operational Feasibility (Operational feasibility includes the process and algorithm; the system will go through to solve the problem and perform its operation.) 3.9.3 Legal & Ethical Feasibility (e.g. The proposed system is legally and ethically feasible as: It does not break any rule and regulation of state. It is purely designed for the assistance of people, so, there is no way that it could harm them. Components data is secure and can be used only by the systems components.
3.10	Summary
Chapter 4	**System Modelling**
4.1	Introduction
4.2	System design
4.3	Design Approach (Generally, there are two basic design approaches in software engineering.) Top Down Design Approach Bottom-Up Design Approach You need to mention about which approach you have adopted to develop the system.
4.4	Interface design 4.4.1 High fidelity prototype (Insert the mockups of your project / screenshots of user interface)
4.5	4+1 view Model of Architecture (give all views and their diagrams which are applicable to your project)

	4.5.1 Logical view (Class diagram) 4.5.2 Process view (Activity diagram, state diagram, sequence diagram) 4.5.3 Development view (Component diagram) 4.5.4 Physical view (Deployment diagram)
4.6	Entity relationship diagram
4.7	Summary
Chapter 5	**Implementation**
5.1	Introduction
5.2	Modules of your FYP (Module by module specify algorithms used and implementation details; include details of any library/framework/ API/ service whatever used.)
5.3	H/W module details (If applicable)
5.4	Summary
Chapter 6	**Result/Testing, Analysis and Validation**
6.1	Introduction
6.2	It is the most important part of your work. You are responsible to test/ validate all your results/achievements in a scientific manner. Be specific and avoid using general terms (e.g. very efficient or user friendly). Achievements are briefly highlighted. Explain in detail your testing setup/arrangements and results. Remember that in scientific work 100% results are not expected or achieved. (Multiple sections are possible)
6.3	Summary
Chapter 7	**Conclusion and Future Work**
7.1	Introduction
7.2	Here the complete project is briefly reviewed and compared with the proposed objectives. Achievements are briefly highlighted. Limitations / claims / future recommendations extracted out of one year's work are to be given. Do not use generalized statements like "there is always room for improvements". (Multiple sections are possible)
7.3	Summary